2.1 Lines and Angles

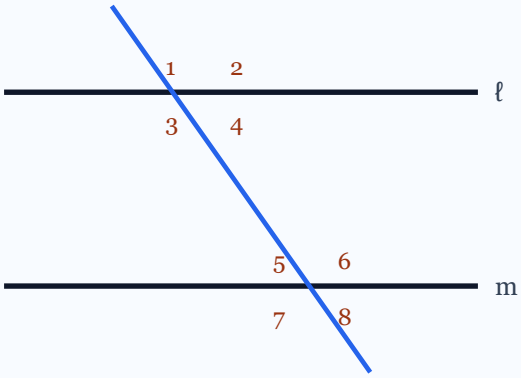
A transversal crossing two lines creates eight angles, and the names — corresponding, alternate interior, alternate exterior, same-side interior — describe exact positional relationships, not vague location. Learning to read these positions correctly (interior vs. exterior, same side vs. opposite side of the transversal) is what makes Section 2.2's parallel-line theorems almost automatic once you get there. This section pushes past simple recall by asking you to classify a pair from a description, count angles across multiple lines, and work out every relationship a single angle has with all seven others.

Examples

Example 1: Setting up the figure

Line t is a transversal crossing lines ℓ and m, creating angles 1–8 as shown. List which angles are interior and which are exterior.

Solution



Transversal t crosses lines ℓ and m, forming 8 angles.

A transversal is a line that crosses two or more coplanar lines at distinct points — here that's t, crossing ℓ and m.

Interior angles lie between ℓ and m: at the ℓ crossing those are $\angle 3$, $\angle 4$; at the m crossing, $\angle 5$, $\angle 6$.

Exterior angles lie outside the two lines: $\angle 1$, $\angle 2$ (above ℓ) and $\angle 7$, $\angle 8$ (below m).

Answer: Interior: $\angle 3$, $\angle 4$, $\angle 5$, $\angle 6$; Exterior: $\angle 1$, $\angle 2$, $\angle 7$, $\angle 8$.

Example 2: Corresponding angles

Explain what makes two angles "corresponding," and name all four corresponding pairs in the figure.

Solution

Corresponding angles occupy the *same relative position* at each intersection — same side of the transversal, and the same rotational corner, just at a different crossing.

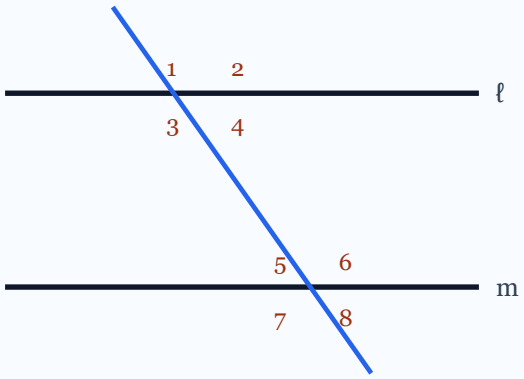
Matching corners: top-left↔top-left, top-right↔top-right, bottom-left↔bottom-left, bottom-right↔bottom-right.

Answer: $(\angle 1, \angle 5)$, $(\angle 2, \angle 6)$, $(\angle 3, \angle 7)$, $(\angle 4, \angle 8)$.

Example 3: Alternate interior angles

Which two pairs of angles are alternate interior angles? State the two conditions they must satisfy.

Solution



∠3, ∠4 are interior at ℓ; ∠5, ∠6 are interior at m.

Two conditions must both hold: (1) both angles are interior (between ℓ and m), and (2) they lie on *opposite* sides of the transversal.

Interior angles are $\angle 3$, $\angle 4$ (at ℓ) and $\angle 5$, $\angle 6$ (at m); pairing opposite sides gives $\angle 3$ (left) with $\angle 6$ (right), and $\angle 4$ (right) with $\angle 5$ (left).

Answer: $(\angle 3, \angle 6)$ and $(\angle 4, \angle 5)$.

Example 4: Same-side interior angles

Which two pairs of angles are same-side (consecutive) interior angles?

Solution

Same two "interior" angles as before, but now paired on the *same* side of the transversal instead of opposite sides.

$\angle 3$ (left, at ℓ) pairs with $\angle 5$ (left, at m); $\angle 4$ (right, at ℓ) pairs with $\angle 6$ (right, at m).

Answer: $(\angle 3, \angle 5)$ and $(\angle 4, \angle 6)$.

Example 5: Alternate exterior angles

Which two pairs of angles are alternate exterior angles?

Solution

Exterior angles are $\angle 1, \angle 2$ (above ℓ) and $\angle 7, \angle 8$ (below m); alternate means opposite sides of the transversal.

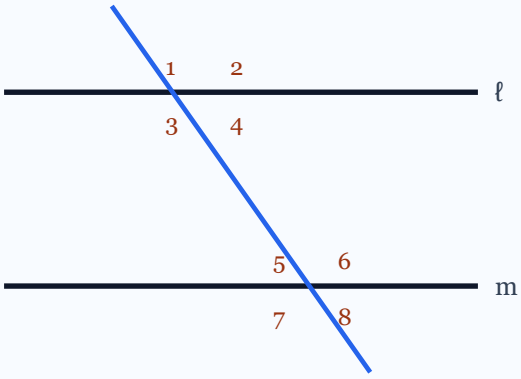
$\angle 1$ (left, exterior) pairs with $\angle 8$ (right, exterior); $\angle 2$ (right, exterior) pairs with $\angle 7$ (left, exterior).

Answer: $(\angle 1, \angle 8)$ and $(\angle 2, \angle 7)$.

Example 6: Vertical and linear pairs at one intersection

At the intersection of t and ℓ alone, name the vertical pair(s) and all linear pairs among $\angle 1, \angle 2, \angle 3, \angle 4$.

Solution



Two lines crossing always make 2 vertical pairs and 4 linear pairs.

At any single intersection of two lines, four angles are formed, giving 2 pairs directly across from each other (vertical) and 4 adjacent pairs (linear pairs).

Vertical: $\angle 1 \& \angle 4$, and $\angle 2 \& \angle 3$.

Linear pairs (adjacent, sum to $180°$): $(\angle 1, \angle 2)$, $(\angle 2, \angle 4)$, $(\angle 4, \angle 3)$, $(\angle 3, \angle 1)$.

Note: this is true at *any* intersection of two lines — the transversal isn't even needed for this fact.

Example 7: Counting angles with more lines

A transversal crosses 3 distinct parallel lines (not just 2). How many angles are formed in total, and how many at each intersection?

Solution

Each place the transversal meets one of the lines is a single intersection of two lines, and two crossing lines always create 4 angles.

With 3 lines being crossed, there are 3 separate intersections.

Total angles $= 4 \times 3 = 12$.

Answer: 12 angles total — 4 at each of the 3 intersections.

Example 8: Interior vs. exterior by coordinates

Horizontal lines $\ell : y = 6$ and $m : y = 2$ are cut by a transversal. A point on the transversal has y-coordinate 4. Is the angle at that point interior or exterior? What about a point with $y = 8$?

Solution

Interior means between the two lines: since $2 < 4 < 6$, the point at $y = 4$ sits between m and ℓ — **interior**.

Since $8 > 6$, the point at $y = 8$ is above both lines, outside the strip between them — **exterior**.

Answer: $y = 4$: interior; $y = 8$: exterior.

Example 9: Classify from a description

$\angle 4$ and $\angle 5$ are both interior angles, and they lie on the same side of transversal t. What are they called, and what will Section 2.2 say about them if $\ell \parallel m$?

Solution

Both interior + same side of the transversal matches exactly one name: **same-side interior angles** (also called co-interior or consecutive interior angles).

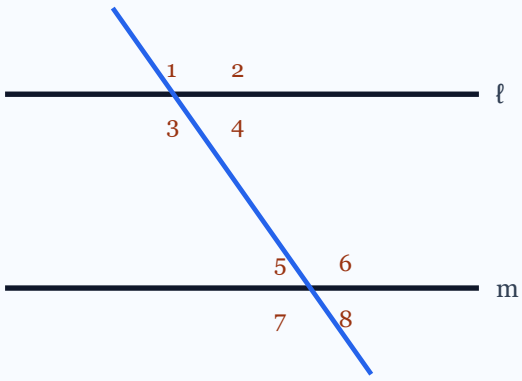
Preview: once $\ell \parallel m$ is established, Section 2.2 proves that same-side interior angles are *supplementary* — unlike corresponding or alternate pairs, which turn out to be congruent instead.

Answer: Same-side interior angles.

Example 10: All seven relationships of one angle

For ∠2 in the figure, classify its relationship to each of ∠1, ∠3, ∠4, ∠5, ∠6, ∠7, ∠8.

Solution



Work outward from ∠2 to every other numbered angle.

Go through each angle relative to ∠2 (top-right, exterior): ∠1 is adjacent at ℓ with outer sides forming a line — **linear pair**. ∠3 is directly across at ℓ — **vertical**. ∠4 is the other angle adjacent to ∠2 at ℓ — **linear pair**.

∠5 is on the opposite side of *t* from ∠2 *and* a different interior/exterior status — **no standard named relationship**. ∠6 is the same corner position at the other intersection — **corresponding**. ∠7 is exterior, opposite side of *t* — **alternate exterior**. ∠8 is exterior, same side of *t* — **same-side exterior**.

Answer: linear pair: ∠1, ∠4; vertical: ∠3; corresponding: ∠6; alternate exterior: ∠7; same-side exterior: ∠8; no special relationship: ∠5.

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Your Turn — Show Your Work!

For each problem, solve it and verify your solution. Show all steps clearly.

Problem 1: Define transversal

A line intersecting two coplanar lines at different points is a what?

Answer: **Verification:**

Problem 2: Corresponding

Matching-position angles at two intersections are called what?

Answer: **Verification:**

Problem 3: Alternate interior

Interior angles on opposite sides of the transversal?

Answer: **Verification:**

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Problem 4: Same-side interior

Interior angles on the same side of the transversal?

Answer: **Verification:**

Problem 5: Alternate exterior

Exterior angles on opposite sides of the transversal?

Answer: **Verification:**

Problem 6: How many angles

Two lines cut by a transversal form how many angles?

Answer: **Verification:**

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Problem 7: Interior location

Interior angles lie where?

Answer: **Verification:**

Problem 8: Exterior location

Exterior angles lie where?

Answer: **Verification:**

Problem 9: Vertical at intersection

At one intersection, opposite angles are what?

Answer: **Verification:**

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Problem 10: Linear pair

Adjacent angles on a straight line at an intersection form a what?

Answer: **Verification:**

Problem 11: Name the pair

Top-left at first intersection and top-left at second: what pair?

Answer: **Verification:**

Problem 12: Name the pair

Bottom-left interior and top-right interior: what pair?

Answer: **Verification:**

Problem 13: Coplanar

Must the two lines cut by a transversal be coplanar for these angle pairs?

Answer: **Verification:**

Problem 14: Challenge — three lines

A single transversal crosses 4 distinct parallel lines. How many intersection points and how many total angles are formed?

Answer: **Verification:**

Problem 15: Challenge — full classification

In the standard 8-angle figure, classify $\angle 6$'s relationship to $\angle 1$, and separately to $\angle 4$.

Answer: **Verification:**

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✓ Answer Key

1. Transversal
2. Corresponding
3. Alternate interior
4. Same-side interior
5. Alternate exterior
6. 8
7. Between the two lines
8. Outside the two lines
9. Vertical
10. Linear pair
11. Corresponding
12. Alternate interior
13. Yes
14. 4 intersection points; 16 angles total
15. $\angle 6$ & $\angle 1$: no special named relationship; $\angle 6$ & $\angle 4$: same-side interior angles

Great work — keep practising!